



Sales and marketing

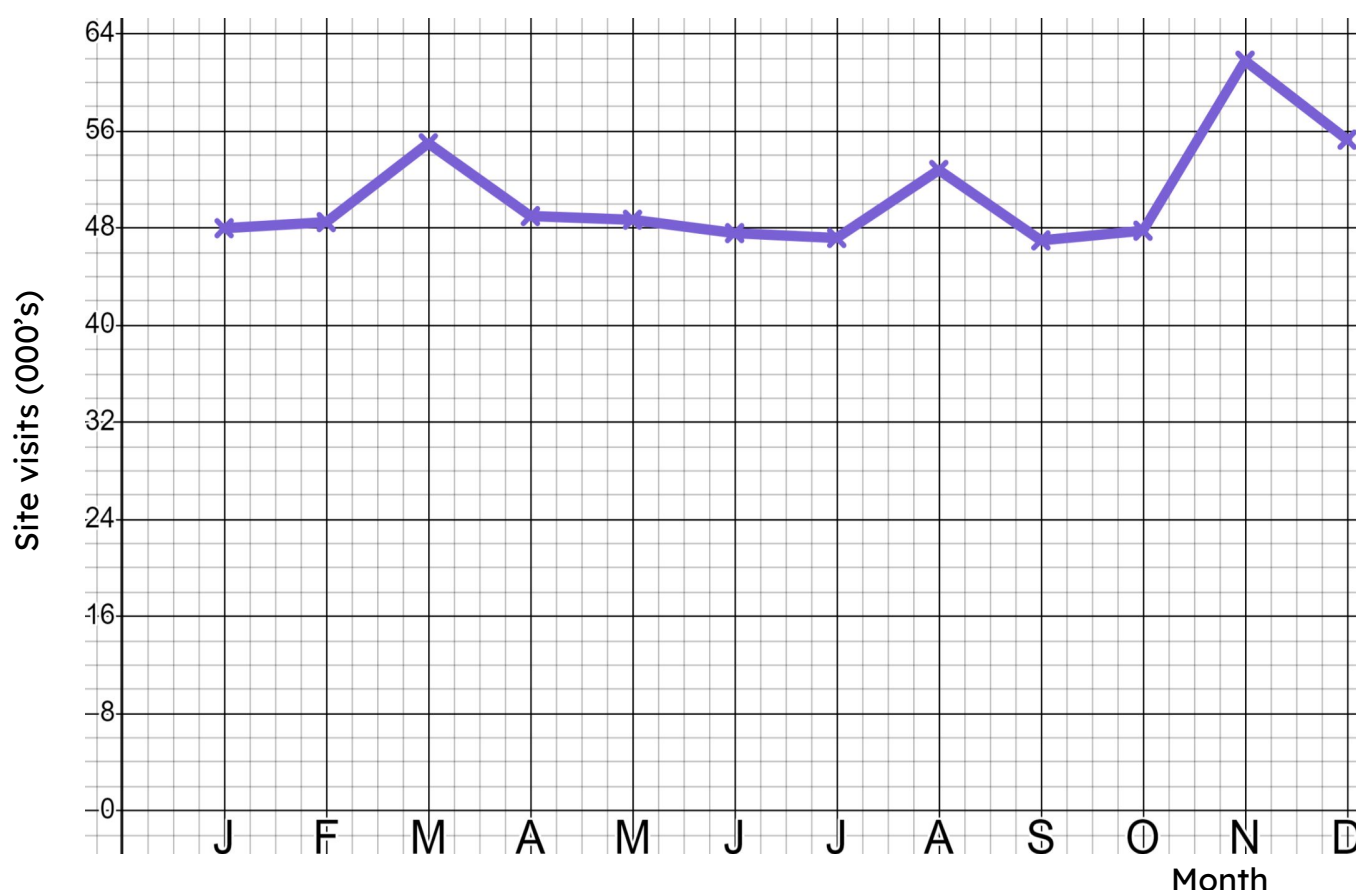
Task A: Maths in marketing

1) 'Sam's Sleepwear' is now well established in the bedwear marketplace. On three occasions in this year they spent just under £5000 on online advertising campaigns.

When were those campaigns?

Which one was the most effective?

How do you know?



You might have said:

"The advertising campaigns happened in March, August and November."

"We know that the November campaign was the most effective one because it resulted in the greatest 'spike' (increase) in site visits."

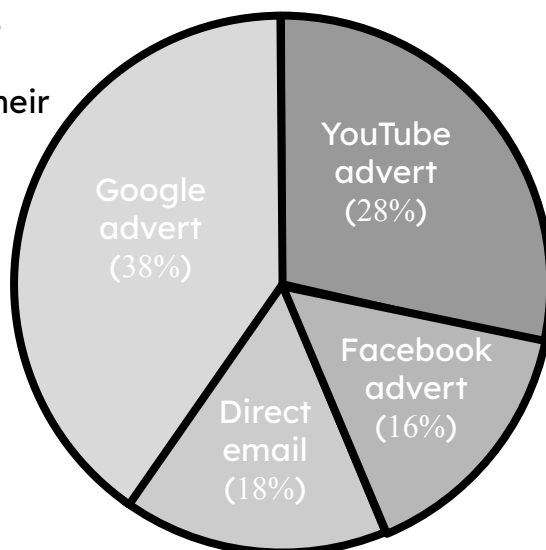
2) 'Sam's Sleepwear' is now well established in the bedwear marketplace. They spend the below on advertising and generate 14 000 extra visitors to their website.

YouTube £1100
Direct email £350

Facebook £500
Google £2500

They now make a sale from 1 in 40 visitors and an average of £14 per sale.

Calculate which form of media is the most **profitable** and advise Sam as to where to spend their next advertising budget.



YouTube £1100

$$0.28 \times 14\,000 = 3920 \text{ visitors}$$

$$£1100 \div 3920 = £0.28... \text{ per visit}$$

$$£0.28... \times 40 = £11.22... \text{ per sale}$$

$$£14 - £11.22... \approx £2.78 \text{ profit}$$

Direct email £350

$$0.18 \times 14\,000 = 2520 \text{ visitors}$$

$$£350 \div 2520 = £0.13... \text{ per visit}$$

$$£0.13... \times 40 = £5.55... \text{ per sale}$$

$$£14 - £5.55... \approx £8.44 \text{ profit}$$

Facebook £500

$$0.16 \times 14\,000 = 2240 \text{ visitors}$$

$$£500 \div 2240 = £0.22... \text{ per visit}$$

$$£0.22... \times 40 = £8.92... \text{ per sale}$$

$$£14 - £8.92... \approx £5.07 \text{ profit}$$

Google £2500

$$0.38 \times 14\,000 = 5320 \text{ visitors}$$

$$£2500 \div 5320 = £0.46... \text{ per visit}$$

$$£0.46... \times 40 = £18.79... \text{ per sale}$$

$$£14 - £18.79... \approx £4.80 \text{ loss}$$

You might have said:

*"Google is expensive and currently **unprofitable** for a business of your size."*

*"Direct email campaigns remain the most **profitable** form of advertising for your business, your market."*

"You spent more than seven times £350 advertising on Google. Perhaps redirect that money and do more direct email campaigns."

*"YouTube and Facebook are both more expensive and less **profitable** than direct email but, they still make a **profit** and give your brand/business good exposure so continue to invest in these."*



Task B: Maths in sales

1) Jun is CEO of a sweet manufacturer that sold 36 000 packets of sweets last year at £1.80 a packet. Their manufacturing cost is £0.60 per unit.



a) Calculate Jun's current revenue and **profit**.

36 000 packets

Selling price: £1.80

Cost price: £0.60

Revenue: $36\,000 \times £1.80 = £64\,800$

Costs: $36\,000 \times £0.60 = £21\,600$

Profit = $£64\,800 - £21\,600 = £43\,200$

b) If Jun could double his sales by halving his price should he?

72 000 packets

Selling price: £0.90

Cost price: £0.60

Revenue: $72\,000 \times £0.90 = £64\,800$

Costs: $72\,000 \times £0.60 = £43\,200$

Profit = $£64\,800 - £43\,200 = £21\,600$

*You might have said, "Whilst this will increase volume and market share it halves your **profit** so I recommend you don't".*

c) If Jun could increase his sales by a third by reducing his price by one third should he?

48 000 packets

Selling price: £1.20

Cost price: £0.60

Revenue: $48\,000 \times £1.20 = £57\,600$

Costs: $48\,000 \times £0.60 = £28\,800$

Profit = $£57\,600 - £28\,800 = £28\,800$

*You might have said, "Whilst this will increase volume and market share it will reduce **profit** by one third so I recommend you don't".*



2) Andeep is a commercial accountant for a guitar manufacturer that sells 1200 units a year at £600 per unit and costs £480 per unit to make. Which of the below are good commercial deals?

Current:

1200 guitars

Selling price: £600

Cost price: £480

Revenue: $1200 \times £600 = £720\,000$

Costs: $1200 \times £480 = £576\,000$

Profit: £144 000

a) 5% uplift in sales for a 10% drop in price? ✗

1260 guitars

Selling price: £540

Cost price: £480

Revenue: $1260 \times £540 = £680\,400$

Costs: $1260 \times £480 = £604\,800$

Profit: £75 600

b) 50% uplift in sales for a 15% drop in price? ✗

1800 guitars

Selling price: £510

Cost price: £480

Revenue: $1800 \times £510 = £918\,000$

Costs: $1800 \times £480 = £864\,000$

Profit: £54 000

c) 10% drop in sales for a 5% increase in price? ✓

1080 guitars

Selling price: £630

Cost price: £480

Revenue: $1080 \times £630 = £680\,400$

Costs: $1080 \times £480 = £518\,400$

Profit: £162 000

d) If Andeep were to drop his price by 12% what percent uplift in sales would he need to see to maintain his existing **profit**?

Current total profit: £144 000

$£144\,000 \div £48 = 3000 \text{ units}$ They would need to see a

New price: $£600 \times 0.88 = £528$

$3000 \div 1200 = 2.5 = 250\%$ 150% increase in sales to

New profit per unit: $£528 - £480 = £48$

achieve the same **profit**.

